

The Trawl — Visual Overhaul Spec

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Purpose

This spec raises The Trawl from blocky placeholder geometry to a finished, moody, first-person 3D look, with the crew and the guns as the two biggest upgrades. It is written for Claude Code and should be read alongside the design doc (The Trawl — Arcade Game 2 Design Document, Draft, Sep 29 2026).

What it supersedes. The design doc was written for a top-down pixel-art deck. The game is now first-person 3D, so this spec replaces three parts of it:

- The "Art" paragraph in the Technical plan (top-down pixel deck, parkour sprite pipeline, porthole bezel framing everything).
- The scope rule "Everything stays procedural (no image or audio files)" as it applies to visuals. Section 4 sets the new rule.
- The row for `trawl_art.cpp` ("Procedural pixel art"). It becomes the 3D art module described here.

Everything else in the design doc (systems, numbers, stations, weapons stats, threats, sound) stands unchanged. This spec changes how things look, never how they play.

Shared code. The Trawl shares its engine and assets with Red Tide, and the two are different games. Build renderer features, the character rig, the hand/viewmodel system, and the gun model kit as shared engine code so Red Tide gets them too. Keep Trawl-specific content (oilskins, the trawler, fish, the dark palette) in Trawl files.

How to use it. Work in the phases in section 9. Before each phase, reread the relevant section here. After each phase, render the screenshot set and compare it against the acceptance criteria. When something in this spec is impossible or a bad trade, say so and propose an alternative instead of quietly doing less.

Diagnosis

The current build reads as unfinished because almost every surface is an untextured, flat-shaded primitive under uniform light. The UI, the telegraph ticker, the chart, and the night palette are good and should be kept.

Area	What the screenshots show	Why it reads as unfinished
Crew (Medic in the bots screenshot)	A stack of boxes: a cuboid torso, flat slab arms that stick straight out, a block head with painted-on squares for a face, a hat made of two flat plates	No anatomy, no silhouette, no joints, no hands, no face. It reads as a placeholder, not a person
First-person view	A single orange stick floats in front of the camera for the gaff, the rod, and the dock walk. The rifle is three boxes	No hands or arms. Nothing tells the player they have a body
Rifle	A dark rectangle barrel on a brown rectangle stock	No receiver, trigger guard, sights, bolt or lever, screws, wood grain, or bluing. It is the least convincing object on screen when it should be the most
Lighting	Even, shadowless light; the deck lamp lights nothing around it	Objects float. No contact shadows or ambient occlusion where crates meet the deck
Materials	Flat colour fills; hard black outlines on everything	Wood, steel, canvas, and skin all look like the same plastic
Boat and dock	Big flat panels, a plain cylinder winch, crates as cubes	No planking, trim, rope, rivets, wear, or clutter
Water and sky	A flat dark plane; a black sky with dots	No waves, glints, foam, or moon. The sea should be the star of a fishing game
Fish	A grey lozenge on the water	No species read: fins, colour, eye, and scale pattern are missing

The reference game (bright, daytime, island) is also low-poly. It looks finished because of soft shading, rounded chunky shapes, gradients, real shadows, expressive characters, and visible hands holding tools. Those qualities carry over to a dark game; the brightness does not.

Art direction

The target is "stylized but grounded, dark but readable": a moonlit steam trawler in fog and rain, lit by warm lanterns, with characters that look like people and guns that look like real period firearms.

Tone. Co-op fishing horror. Dread comes from darkness, fog, and the unknown below the hull, never from a muddy or unreadable image. A player should always be able to read the deck, their crewmates, and their own hands. What lives beyond the lantern light can be black.

Era. A steam trawler with a coal boiler, an engine telegraph, shillings, a hardhat diving rig, and lever carbines. Treat it as roughly 1895–1915, a fictional North Atlantic-meets-tropical setting. Clothing, tools, lamps, and guns all come from that period. Nothing modern: no plastic, no zips, no hi-vis, no polymer guns.

Two levels of realism. The game deliberately mixes two levels of detail:

1. **Stylized layer** (people, fish, boat, dock, sea). Simplified forms, slightly exaggerated proportions, clean bevelled shapes, painterly gradients, readable silhouettes. Think a well-made indie game, not a photoscan.
2. **Realistic layer** (guns, and to a lesser degree the knife, gaff, and fishing reels). Accurate proportions, real mechanical parts, true materials (blued steel, case-hardened colour, brass, oiled walnut). The gun in your hands should be the most detailed object on screen.

The contrast is intentional. Guns are the expensive, deliberate, dangerous objects in this world, and the detail sells that.

Palette.

Role	Colours	Notes
Night sea and sky	Deep blue-black, slate, cold teal	Never pure black except far beyond the light
Moonlight (key light)	Pale blue-white, low intensity	Gives every silhouette a cool rim
Lantern and lamp light	Warm amber to orange, 2200–2700 K	The main working light on deck; pools and falls off

Role	Colours	Notes
Oilskins and gear	Faded yellow, tar black, oxblood, bottle green	Weathered, never saturated
Wood	Grey-brown weathered deck, darker varnished trim	Wet areas darker and glossier
Metal	Rusted iron, dull brass, blued gun steel	Brass catches lantern light
Danger and blood	Dark red, near-black under moonlight	Blood should read but not glow

Reference quality bar. Match the finish of the reference screenshots (soft shading, rounded forms, real shadows, expressive characters, visible hands), not their content or brightness. Do not copy their characters, layouts, or assets.

Asset pipeline

Recommendation: keep everything generated by code, but move generation offline into build-time generator programs that write baked mesh and texture files, instead of building primitives at runtime.

The old rule ("no image or audio files") is why the crew are boxes: runtime code that stacks cubes cannot produce a convincing person or rifle. The fix is not hand-modelling. It is better generators.

The new rule.

- Every model and texture is still produced by code Claude Code writes and keeps in the repo, so it can be regenerated and edited by changing code.
- Generators run at build time (a `tools/artgen/` program, or Blender run headless with Python scripts) and output glTF 2.0 meshes and PNG texture sets into `assets/trawl/` and `assets/shared/`.
- The game loads those files. Generated files are committed so the game builds without Blender.
- No downloaded, purchased, or third-party art. No AI image generation.

Why Blender scripting is the strongest option. It gives subdivision, bevel and solidify modifiers, proper UV unwrapping, normal-map baking, ambient-occlusion baking, automatic weights for skinning, and armatures, all from Python. Writing those from

scratch in C++ is weeks of work. If Blender is not acceptable, the fallback is a C++ generator with signed-distance-field modelling (smooth unions of capsules and boxes), marching cubes, and smoothing, plus procedural texture code.

Techniques the generators must use.

- Bevels or rounded edges on every hard edge (2–6 mm on guns, 1–3 cm on props). Light catching edges is most of what makes low-poly look finished.
- Smooth normals across curved surfaces, hard normals only at real creases.
- Baked ambient occlusion into vertex colour or an AO map.
- PBR texture sets (base colour, normal, roughness, metallic, AO) for guns and close-up tools; vertex colour plus a small gradient or detail texture for stylized-layer objects.
- Level-of-detail meshes (LOD0–LOD2) for characters and guns seen in third person.

Budgets.

Asset	Triangles (LOD0)	Textures
First-person gun (viewmodel)	8,000–20,000	2048 ² PBR set
Third-person gun (held by others)	1,500–3,000	512 ² PBR set
Crew character, full body	6,000–12,000	1024 ² body + 512 ² head
First-person arms	4,000–8,000	1024 ²
Fish (per species)	1,000–4,000	512 ²
Deck props	200–2,000	Shared trim sheet

This needs a decision from you before Claude Code starts phase 1. Open question: is Blender (free, run headless by scripts) acceptable as a build-time tool?

Characters: crew and bots

Every crewman, human or bot, must read as a weathered working sailor from a few metres away in lantern light: a real body, a face with an expression, two working hands, and clothes that say what job they do. Players and bots use the same character system; bots are not a lower-detail version.

Body and proportions

Build one skinned base body, not a stack of separate boxes. The body is a single continuous mesh (or a few meshes that overlap cleanly at the collar, cuffs, and boot tops) so joints bend without gaps.

- **Proportions.** Stylized-realistic, about 7 heads tall. Slightly enlarged hands (about 1.15×) and head (about 1.1×) so they read at distance and in first person. Broad, sloped shoulders; a working man's build.
- **Forms.** Rounded and tapered, never boxy. Arms taper from shoulder to wrist with a visible elbow. Legs have knees. The torso has a chest, a waist, and a slight belly. Necks exist.
- **Body variants.** Generate at least 4 builds from the same rig by blend shapes or generator parameters: lean, stocky, tall-rangy, heavysset. Plus height variation of $\pm 6\%$.

Faces

The painted-square face is the single biggest problem. Replace it with a sculpted head.

- Real facial structure: brow ridge, eye sockets, a nose with a bridge and nostrils, cheekbones, a mouth with lips, a jaw, chin, and ears.
- Eyes are separate meshes (eyeball with iris and a wet specular highlight) inside sockets, with eyelids that blink every 3–6 s.
- Stylized, not realistic: slightly larger eyes, simplified planes, clean shapes. Aim between a good animated film and a painted portrait, never photoreal (that falls into the uncanny valley at this budget).
- At least 8 head variants by generator parameters: face width, nose shape, brow height, jaw, age lines, and ear size.
- Facial hair as geometry cards or sculpted shells: full beard, mutton chops, moustache, stubble (a texture), clean-shaven. Hair under hats as short sculpted shells.
- Skin tones: a range of at least 6, weathered and wind-reddened at the nose and cheeks.
- Expressions as blend shapes: neutral, strain (reeling, hauling), fear (threat nearby), pain (injured), shout (barks and voice), and grin (a big catch). Voice chat drives a simple jaw-and-lip flap from audio amplitude.

Hands

Hands matter more than anything else on the body, because the player sees their own in first person all game.

- Five fingers, each with three bending segments, knuckles, fingernails, and a thumb that opposes.
- Skin: weathered, with visible tendons and veins on the back of the hand, rough knuckles, and dirty nails. Optional fingerless wool gloves or leather work gloves per role.
- A hand-pose library: relaxed, fist, grip (rod, gaff, rope), pistol grip with trigger finger, support hand under a fore-end, pinch (bait), flat (pushing), and pointing (ping wheel orders).

Clothing by role

Clothes are separate meshes on top of the body, with thickness (no paper-thin shells), folds at the elbows and knees, seams, buttons, and wear. Each role has a silhouette you can identify at 20 m.

Role	Signature silhouette	Outfit
Bosun	Broad, heavy, short peaked cap	Wool pea coat over a cable-knit gansey, canvas trousers, leather apron, sea boots. Patch kit on the belt. Rolled sleeves show forearms
Angler	Wide sou'wester, long oilskin	Yellow oilskin smock and hat, rubber sea boots, fishing knife on a lanyard, bait tin at the hip
Diver	Bulky, a lamp on the head	Heavy wool sweater, canvas diving dress folded to the waist when on deck, helmet lamp, knife strapped to the calf. In a dive, the full hardhat rig: copper helmet with portholes, corselet, lead boots
Medic	Long coat, bag across the body	Long waxed coat, waistcoat, rolled shirt sleeves, round spectacles, red-cross armband, leather medical satchel

Cosmetic variation within a role: colour (faded yellow, black, oxblood, bottle green oilskins), scarves, knitted caps vs sou'westers, patches, and tattoos on the forearms.

Materials on characters

- **Skin.** Subsurface-like warmth (a wrap-lighting term plus a reddish shadow tint), roughness about 0.5, a slight sheen on the nose and forehead when wet.
- **Oilskin.** Waxy, roughness 0.3–0.45, with cracked and faded areas at folds.
- **Wool.** Fuzzy edges via a fresnel-softened rim, roughness 0.9, a knit normal pattern.
- **Leather.** Scuffed, darker at edges, roughness 0.6.
- **Wetness.** A per-character wetness value (rain, spray, overboard) darkens base colour by about 30% and drops roughness by about 0.3, with drips at the hat brim.
- **Blood and grime.** Decal layers on hands, apron, and sleeves that build up through a night and wash off in rain.

Rig and animation

One humanoid skeleton shared with Red Tide: root, pelvis, three spine joints, neck, head, jaw, eyes, clavicles, upper and lower arms with twist bones, hands with full fingers, upper and lower legs, feet and toes. Animation runs on the client from host state; nothing here changes networking.

Required animation set, procedural or keyframed by generator scripts:

- **Locomotion.** Idle (with breathing and weight shifts), walk, run, crouch, and a sea-legs layer that counter-balances the boat's roll and pitch. At over 12° of roll, the body leans and arms reach out to brace. Slipping on a wet deck gets a stumble.
- **Stations.** Steering at the helm, shovelling coal, pumping, winching, gutting at the table (knife strokes on a fish), casting, hook-setting, reeling with strain, side pressure, gaffing, and hauling the net.
- **Combat.** Each melee weapon's swing set, aiming and firing each gun class, reloading in third person (matches the first-person reload), and recoil, including the nitro express and punt gun knocking the shooter back.
- **Reactions.** Flinch, hit, a broken arm hanging limp (per the design doc), bleeding with a hand pressed to the wound, being grabbed, being charmed (a slack, entranced stare), falling overboard, and swimming or treading water.
- **Communication.** Pointing for ping-wheel orders, waving, a thumbs-up, and a shrug. Head and eyes look at whoever is talking or at the current threat.
- **Death and ghosts.** A ragdoll on death with the body staying where it fell. Ghosts use the same mesh with a pale translucent shader, a cold fresnel rim, drifting hair and clothes, and no shadow.

IK on top of all of this: feet planted on the rolling deck, hands locked to the helm spokes, the rod, and the winch handle, and head look-at.

First-person body

The player has a body, not a floating stick.

- Separate first-person arms (higher detail, their own field of view at 60–70°, no clipping into walls) wearing the sleeves of the player's role outfit.
- Looking down shows the player's own legs and boots.
- Every held item is held by both hands as it should be: the gaff two-handed, the rod with one hand on the reel and one on the grip, the fish priest in a fist, the life ring under an arm, guns as in section 6.
- Subtle idle sway, breathing, walk bob tied to footsteps, and a sea-legs drift tied to the boat's motion. A motion-comfort slider scales all of it.
- Hands react to state: shaking when the tension gauge is in the red, wet and dripping in rain, bloodied after gutting, and the broken arm unable to hold a two-handed item.

Bots specifically

Bots use the full character system. On top of that:

- Each bot gets a generated identity at spawn: a name, body build, head, beard, skin tone, and outfit colours, from a seed so all clients see the same sailor.
- The skill level shows in body language: Green bots fumble (a dropped grip, a slower cast, glancing around), Able bots are steady, Old Hands move with economy and look relaxed.
- When a bot "barks" (subtitled per the design doc), it turns its head and does the shout expression and a gesture.
- The name tag above a bot is small, warm white, and fades beyond 15 m, matching how player names show in the reference.

Weapons

Guns are the most realistic objects in the game: every firearm is modelled part by part on a real period design, with true materials, wear, and fully animated mechanisms. A player should be able to look at the viewmodel and see how the gun works.

Construction standard

Model each gun as an assembly of real parts, not a silhouette. A gun is not finished until it has all the parts its real counterpart shows from the outside.

- **Metal parts.** Barrel(s) with a visible bore and crown at the muzzle, receiver or frame, trigger, trigger guard, hammer or bolt or lever, sights (front blade or bead, rear notch or ladder), screws with slots, pins, springs where visible, sling swivels, and a magazine tube, cylinder, or box as the design calls for.
- **Wood parts.** Stock and fore-end shaped with real ergonomics (comb, wrist, butt, grip), chequering on the grip, and a butt plate (steel or horn).
- **Moving parts are separate objects** with their own pivots so they animate: cylinders rotate, hammers cock, levers swing, bolts slide, break-actions open, triggers pull, ejectors kick.
- **Scale.** True to life in centimetres. A lever carbine is about 95 cm long, a service revolver about 26 cm. Then the viewmodel FOV and position frame it.
- **Bevels on every edge**, 0.5–2 mm. Machined edges catch a thin line of lantern light; this alone makes guns read as real.

Materials

Material	Look	PBR values (approximate)
Blued steel	Deep blue-black, worn to bright silver at edges and high points	Metallic 1.0, roughness 0.25–0.4, blue tint in base colour
Case-hardened steel (receivers, hammers)	Mottled blue, straw, and purple	Metallic 1.0, roughness 0.3, colour from a procedural noise texture
Bright or worn steel	Grey with fine machining marks	Metallic 1.0, roughness 0.35, anisotropic streaks in the normal map
Rust and pitting	Orange-brown speckle in crevices and near the muzzle	Metallic 0.0, roughness 0.9, masked by curvature and AO
Brass	Warm gold, darkened in recesses	Metallic 1.0, roughness 0.3–0.5

Material	Look	PBR values (approximate)
Oiled walnut	Red-brown with visible grain and figure	Metallic 0.0, roughness 0.35–0.5, procedural grain following the stock's length
Chequering and horn	Fine diamond pattern; mottled translucent horn	Normal map; roughness 0.6

Wear is generated, not painted: edge wear from a curvature map, handling wear where hands touch (grip, fore-end, lever), dirt and old oil in crevices from the AO map, and a few dings and scratches. Every gun gets small markings: a maker's stamp, a serial number, proof marks, and a few engraved scrolls on the nicer guns. All are invented, never real trademarks.

Guns respond to the world: wet in rain (beading, lower roughness), blood spatter after deck kills, and a bright specular streak when a lantern or flare passes over them.

Per-weapon reference

Each gun in the design doc's catalogue gets a real period design as its visual reference. Model the look, not the name; markings stay fictional.

Weapon (design doc)	Visual reference	Key visible features to model
Service revolver	British top-break service revolver, about 1900	Top latch, hinged frame, six-chamber fluted cylinder, lanyard ring, bird's-head grip. Breaking it open lifts the ejector star
Derringer	Over-under two-shot pocket pistol	Stacked barrels that hinge up, spur trigger, small hammer, rosewood grips
Pepperbox	Four-barrel rotating pepperbox	Barrel cluster rotates with each pull, bar hammer, ornate frame
Flare pistol	Brass break-open flare pistol	Large bore, brass frame, wooden grip; the flare shell is visible when loading

Weapon (design doc)	Visual reference	Key visible features to model
Speargun and twin speargun	Brass-and-wood spring speargun	Wooden stock, brass muzzle, cocking lever, a steel spear with barbs, a line spool
Harpoon pistol	Fictional, in the same brass-and-wood family	A short barrel holding a barbed harpoon head, a rope coil on the side
Air pistol and air rifle	Reservoir air guns of the period	Brass or iron air reservoir as the butt, a pump lever, a fine rifled barrel
Captain's pistol	Ornate flintlock-era officer's pistol (a relic)	Silver inlay, engraved lock, ghostly patina, a faint green verdigris
Coach shotgun	Side-by-side double, external hammers, short barrels	Two hammers that cock separately, top lever, break-open, brass shells that extract
Lever carbine	1890s-style lever-action saddle carbine	Tube magazine under the barrel, loading gate on the side, saddle ring, barrel band, ladder rear sight. The lever ejects each case
Chatter gun	Fictional, inspired by early automatic guns	Finned cooling shroud, top pan or side drum, spade grip or pistol grip, a visible feed; the drum magazine attachment is a separate model
Long rifle (scoped)	Bolt-action service rifle with a long brass telescopic sight	Bolt handle, box magazine, stripper-clip guide, full-length wood, brass scope with mounts
Nitro express	Side-by-side double rifle, big game	Heavy barrels, express sights with folding leaves, engraved side-locks, huge cartridges
Tribal longbow	Laminated wooden bow	Grip wrap, string, fletched arrows recovered from kills

Weapon (design doc)	Visual reference	Key visible features to model
Punt gun	Single huge-bore wildfowl gun	Barrel taller than a man, a rope breeching, a swivel mount when on the rail
Steam riveter	Fictional industrial tool-weapon	Brass pressure gauge, a steam hose, a riveting head, a rivet hopper
Blunderbuss	Flared-muzzle flintlock	Bell muzzle, brass furniture; junk ammo visible when loading
Galvanic prod	Fictional electrical device	Glass jar cells, copper coils, insulated grip, visible sparks
Rocket harpoon	Fictional	A finned rocket harpoon on a launcher tube, a fuse, a tether line
Harpoon cannon and swivel gun	Deck-mounted whaling cannon; rail swivel gun	Heavy cast mount, aiming handles, a harpoon with a grenade head; bronze swivel gun on a yoke

Melee weapons are in the realistic layer too, at slightly less detail: a knife with a riveted wooden handle and a honed edge, the gaff with a forged steel hook and a worn ash shaft, the fish priest as a turned and lead-weighted wooden club, brass knuckles, a cutlass with a basket guard, a cleaver, a boat hook, a marlinspike, and so on.

Attachments and upgrades are visible

- Every attachment in the design doc is its own model mounted on the gun: sights, compensators, a choke at the muzzle, a bayonet with a lug, a sling with leather straps, an oilskin cover over the breech, a baffle (a period can-type silencer), a bone stock replacing the wood, a brass night glass, a lodestone sight with a dark stone set in brass, and drum or extended magazines.
- Damage upgrades show as finish tiers: tier 1 cleaned and oiled, tier 2 new rifling with a polished bore and fresh bluing, tier 3 engraved scrollwork and gold-inlaid lines.

First-person viewmodels

- Both hands hold the gun correctly: firing hand on the grip with the trigger finger on the trigger, support hand under the fore-end. Fingers wrap around the wood; no floating hands.
- Position the gun lower-right and angled so the action, sights, and the player's hands are all visible. Aiming down sights brings the actual iron sights or scope to the eye, aligned.
- Use a separate viewmodel FOV (around 55–65°) and depth range so guns never clip into walls or distort.
- A two-handed long gun cannot be fired with a broken arm; the viewmodel shows the arm hanging and the gun held one-handed and low.

Animation and effects

Each gun gets a full animation set: draw, holster, idle sway, aim, fire, dry fire, reload (partial and from empty), inspect, and clearing a misfire (the design doc's wet-weather misfires).

Reloads are mechanically true, step by step: the revolver breaks open, empties, and takes rounds one at a time (or all six with a speed loader). The lever carbine takes rounds through the side gate one by one, and can be interrupted. The double guns open, extract, and drop in two shells. The bolt rifle works the bolt and pushes rounds from a stripper clip.

On firing:

- Recoil moves the gun and the camera by weapon class. The nitro express and punt gun knock the shooter back, as in the design doc.
- The hammer falls, the action cycles, and cases eject with physics and bounce on the deck with a sound.
- A muzzle flash lights the surroundings for one or two frames, a real point light that throws shadows. Black-powder guns (blunderbuss, punt gun, flintlocks) leave a thick white smoke cloud that drifts with the wind; smokeless guns leave a thin wisp.
- Hits on fish spray blood and scales; misses splash on the water surface; rounds that hit the water show the slowdown from the design doc's ballistics.
- Damage tiers, attachments, and wear all show on the third-person model held by other crew, at lower detail.

Lighting, rendering, and post

The renderer moves to a physically based forward+ (or deferred) pipeline with real shadows, ambient occlusion, HDR, and tonemapping, all in the shared engine. Darkness comes from light design, not from turning everything down.

Shading model. PBR (metallic-roughness, GGX specular, energy-conserving diffuse) for everything. Characters add a wrap-diffuse term for soft skin and cloth. The hard black outlines are removed by default; keep a thin, colour-tinted outline as an option to test, at most 1 px and never pure black.

Lights.

- **Moon (key).** Directional, pale blue, low intensity, with cascaded shadow maps (3–4 cascades, soft PCF or PCSS filtering). Its direction and strength follow the design doc's moon phase; a waning moon is dimmer.
- **Practical lights.** The masthead lamp, wheelhouse lamp, lantern mast (with its filter colour), deck lanterns, dock lamps, the helmet lamp, and the boiler's fire door are real point or spot lights with shadows, inverse-square falloff, and warm colour. The lantern mast level in the design doc changes their actual intensity and radius.
- **Muzzle flashes, flares, and lightning** are short-lived lights that cast shadows.
- **Ambient.** Hemisphere ambient (cool sky above, very dark sea below) plus light probes on the boat, so shaded sides are dim blue rather than black.

Ambient occlusion. Screen-space AO (GTAO or HBAO) on top of the baked AO from the generators. Contact shadows where boots, crates, and fish touch the deck are essential.

Fog and atmosphere. Exponential height fog with in-scattering from the moon and from each lantern (volumetric or a cheap screen-space approximation), so lanterns get a glow halo in fog. Density follows the night's weather: lagoon fog thick, rain medium, clear nights thin.

Post-processing. HDR rendering, ACES-style tonemapping, bloom only on bright sources (lamps, flashes, eye glints, phosphor sonar), a subtle vignette, film grain at a low level, and a colour grade per ground and per weather (a cold teal-blue night, warm lantern highlights). Optional TAA or SMAA for clean edges.

Readability rules.

- Crewmates always keep a faint moonlit rim so they read against the dark sea.
- The area within about 6 m of an active station lantern is fully readable.

- Anything that matters for gameplay (a threat tell, a fish breaking the surface, a dropped item) is never fully black when it is meant to be seen.
- Provide a brightness or gamma calibration screen in settings.

World: water, sky, boat, and fish

The sea, the trawler, and the catch get the stylized-layer treatment: clean shapes, rich surface detail, and lighting that makes them feel wet, heavy, and alive.

Water.

- A sum of Gerstner waves (or an FFT ocean) whose height matches the boat physics in `trawl_boat.cpp`, so the hull visibly rides the waves it is simulated on.
- Fresnel reflection of the sky and moon, a moon glitter path, and reflections of the boat's lanterns stretched across the swell.
- Foam at the bow, along the hull, in the wake behind the screw, where lines enter the water, and where fish thrash. Blood from the scuppers spreads as a dark stain on the surface, tying into the design doc's blood mechanic.
- Below the surface: darkness that fades with depth, light shafts under lanterns, and silhouettes of fish inside the light radius (matching what the network sends per the design doc).
- Rain rings on the surface in rain.

Sky and weather.

- A sky dome with a gradient, a moon whose phase matches the telegraph, clouds lit from behind by the moon, and stars hidden by fog and cloud.
- Rain as streaks lit by nearby lanterns, splashes on the deck and rail, drips from rigging and hat brims, and wet surfaces (darker, glossier) everywhere it falls.
- Fog as described in section 7, with drifting banks rather than a uniform wall.
- Lightning (when weather calls for it) lights the whole scene for a frame.

The boat.

- Rebuild the trawler from planks and frames: individual deck planks with caulked seams, bulwarks with a cap rail, scuppers, hatch coamings, a wheelhouse with window frames and glass, a funnel with soot, and rigging and stays as real lines.
- Detail every station from the design doc as a real machine: a brass-fitted helm, an engine telegraph with a needle and bell, a steam winch with gears and drums, a hand pump with a lever, the gutting table with a cutting board, scale buckets and knives, the

boiler with a fire door that glows, the net winch, the cannon mount, and the diving gantry.

- Clutter that tells a story: coiled rope, nets, fish boxes, ice, buckets, oilskins on hooks, a tea kettle, bait tins, empty shells, and blood on the planks after a kill.
- Wear: salt stains, tar, rust streaks under iron fittings, and scuffed paint.
- The dock: planks on piles with barnacles, warehouse sheds with corrugated roofs, gas lamps, the Gunsmith's shop with racked guns and a counter, crates, and a crane.

Fish.

- Each species from the design doc's rosters gets its own model with recognisable anatomy: fins in the right places, gill plates, a lateral line, an eye with a wet highlight, and a mouth that opens.
- Iridescent scales (a thin-film or angle-dependent colour shift) and a slime sheen while fresh. They dull after death and cleaning.
- Rigged with a spine chain for swimming, flopping on deck, and thrashing on the line; procedural animation is fine.
- Gutting and cleaning change the model: cut open, then a clean fillet look on the table.
- Glimmer variants get a pale, faintly glowing colouring per the design doc.
- Threats and big fish (shark, marlin, sturgeon, the Kraken, the White) get the most effort, with scars, distinct tells, and heavy, slow animation that sells their size.

Process and acceptance

Work in seven phases, each ending with a before/after screenshot set that Ray approves before the next phase starts.

Screenshot harness. Extend the design doc's `--trawl-shots` self-test so it also renders a fixed set of 3D camera shots to PNG, before and after each phase, at 1920×1080:

1. Helm view at night in lagoon fog (matches the current helm screenshot).
2. Deck view of a crewman at the gutting table in rain (matches the current bots screenshot).
3. The dock at night (matches the current dock screenshot).
4. First person aiming the lever carbine at a fish on the surface in rain.
5. First person on the starboard rod mid-fight.
6. A character turnaround: each role, front, side, back, and a face close-up, under lantern light.

7. A gun turnaround: each gun, side and three-quarter views, on a neutral lit background, plus the viewmodel at rest and aiming down sights.
8. All four bots side by side, to check that they look like different people.

Claude Code looks at its own renders, compares them with the reference images and this spec, and fixes problems before showing Ray.

Phases.

Phase	Work	Done when
1. Pipeline	Generator tooling (section 4), glTF loading, PBR materials, the screenshot harness	A test gun and a test head load in-game with full PBR materials
2. Lighting	Shadows, AO, practical lights, fog, tonemapping, post (section 7)	Shots 1–3 look moody and readable with no art changes yet
3. Characters	Base body, heads, hands, role outfits, rig, core animations, bots' identities (section 5)	Shots 2, 6, and 8 pass the character criteria below
4. First-person	Arms, hand poses, held items, body awareness	Shots 4 and 5 show real hands holding real tools
5. Guns	Every gun, attachments, upgrade tiers, viewmodels, animations, effects (section 6)	Shot 7 passes the gun criteria below
6. World	Water, sky, weather, boat, dock, fish (section 8)	All shots look finished; nothing left as a raw primitive
7. Polish	Performance, LODs, settings, bugs	Holds the frame-rate target with six players

Character criteria.

- No visible boxes, slabs, or flat painted faces on any crewman.
- Every role can be named from its silhouette alone at 20 m.
- Faces show eyes with highlights, a nose, a mouth, and a readable expression.
- Hands have five bending fingers that grip what they hold.
- Four bots side by side look like four different people.
- Joints bend without gaps or tearing at the shoulders, elbows, hips, and knees.

Gun criteria.

- Each gun matches its reference in the per-weapon table, with all its external parts.
- Every moving part animates on reload and firing.
- Materials read as blued steel, case-hardened steel, brass, and walnut under lantern light, with edge wear.
- In a side-by-side screenshot, the gun is visibly the most detailed object in frame.
- Iron sights and scopes line up when aiming down sights.

Performance target. 60 fps at 1080p on a mid-range PC (around a GTX 1660 or RX 5600) with six crew on deck in rain and fog. Settings expose shadow quality, AO, fog quality, and resolution scale. If a feature in this spec costs too much, flag it with numbers and propose a cheaper version.

Open questions for Ray.

- Is Blender acceptable as a headless build-time tool (section 4)?
- Should the hard outlines be removed entirely, or kept thin as part of the game's identity?
- What is the actual target PC?

Kickoff prompt for Claude Code

Paste this into Claude Code with the doc link and the screenshots (current game plus the two references) attached.

Read "The Trawl – Visual Overhaul Spec" in full before doing anything. It replaces the Art paragraph, the pixel-art scope rule, and the trawl_art.cpp description in The Trawl design doc; every other part of the design doc still stands. It changes how the game looks, never how it plays.

Attached: screenshots of the current build and two reference images. Match the references' level of finish, not their brightness or content. The Trawl stays a dark night game.

The two biggest upgrades:

1. Crew and bots must look like real, weathered sailors with faces, hands,

role outfits,
and full animation (spec section 5), not stacked boxes.
2. Guns must be the most realistic objects in the game: real period designs,
part by part,
true materials and wear, fully animated mechanisms (spec section 6).

Build renderer, rig, viewmodel, and gun-kit code as shared engine code so Red Tide benefits.

Start with phase 1 only. First, answer the spec's open questions or tell me what you need from me, then build the pipeline and the screenshot harness. At the end of each phase, show me before/after renders from the harness, check them against the spec's acceptance criteria yourself, and wait for my approval before starting the next phase. If anything in the spec is impossible or too expensive, say so with numbers and propose an alternative instead of quietly doing less.